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Development of a novel rotary magnetic refrigerator☆

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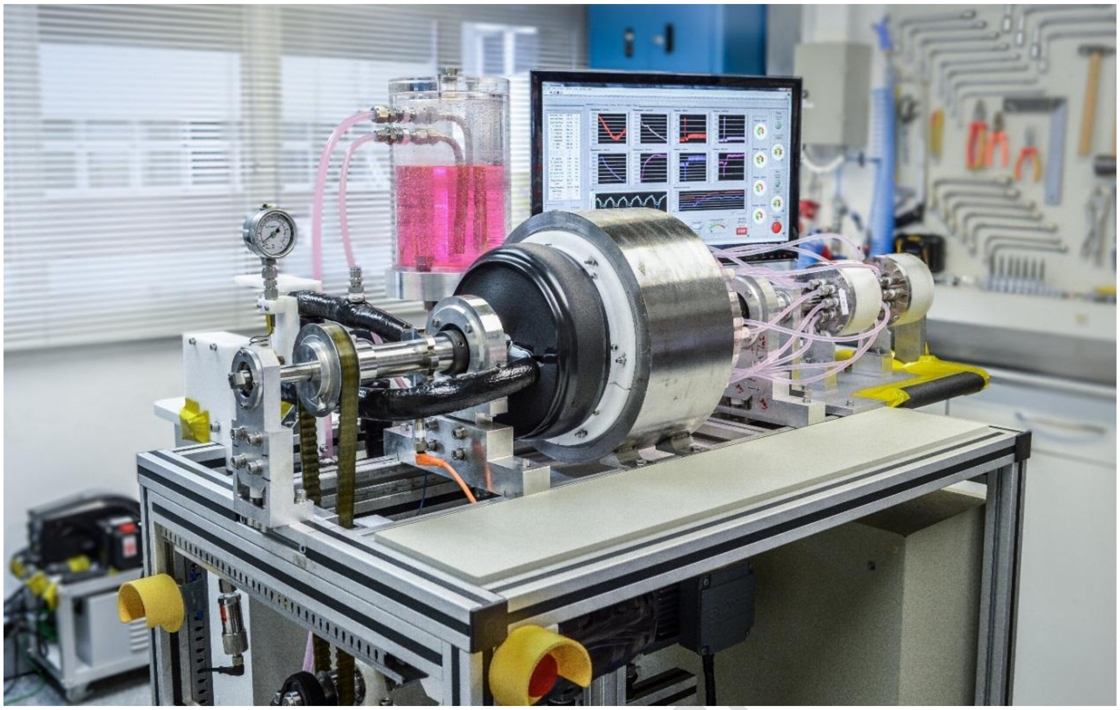
HIGHLIGHTS

- Design aspects and performance data of a novel rotary-type magnetic refrigerator
- New fluid flow distribution system placed at the hot end to reduce frictional heating
- Experimental results for different conditions of thermal load, frequency and flow rate
- System performance evaluated in terms of the COP and the second-law efficiency
- The device generated a temperature span of 7.1 K for a thermal load of 80.4 W

Graphical Abstract



Photograph of the novel rotary magnetic refrigerator developed at POLO/UFSC.

Abstract

A novel rotary magnetic refrigerator was designed and built at the Federal University of Santa Catarina (UFSC). The optimized magnetic circuit is a two-pole system in a rotor-stator configuration with high flux density regions of approximately 1 T. Eight pairs of stationary regenerator beds filled with approximately 1.7 kg of gadolinium spheres (425-600 μm diameter) were placed in the magnetic gap. Two low-friction rotary valves were developed to synchronize the hydraulic and magnetic cycles. The valves were positioned at the hot end to avoid heat generation in the cold end. In this work, experimental results are presented as a function of the operating frequency, fluid flow rate, hot reservoir temperature and thermal load. The performance of the device was evaluated in terms of the coefficient of performance (COP) and overall second-law efficiency (η_{2nd}). The maximum no-load temperature span was 12 K at 1.5 Hz and 150 L h^{-1}, and the maximum zero-span cooling power was 150 W at 0.8 Hz and 200 L h^{-1}. For a thermal load of 80.4 W, at 0.8 Hz and 200 L h^{-1}, the device generated a temperature span of 7.1 K, with a COP of 0.54 and η_{2nd} of 1.16%.

Keywords:

Magnetic refrigeration, magnetocaloric effect, permanent magnet, regenerator, gadolinium, coefficient of performance

Nomenclature

Roman

A	Area	m^2
B	Magnetic induction	T
c	Specific heat	J kg^{-1} K^{-1}
f	Frequency	Hz
H	Magnetic field	A m^{-1}
L	Length	mm
m	Mass	kg
N	Number	
p	Pressure	bar
R	Radius	mm
T	Temperature	K
V	Volume	m^3
w	Width	mm

Greek

Δ	Change	
η	Eficiency	
Λ_{cool}	Figure of merit for magnetic circuits for M.R.	$T^{2/3}$
ϕ	Utilization factor	
ε	Porosity	
σ	Standard deviation	
τ	Blow period	

Composed and constants

A_c	Cross sectional area	m^2
BH	Energy product	kJ m^{-3}
HC	Magnetic coercivity	kA m^{-1}
μ_0	Magnetic permeability of	$4\pi \times 10^{-7}$ N

	vaccum	A^{-2}
μ_r	Relative magnetic permeability	
$\dot{W}$	Power	W
Δp_{sys}	System pressure drop	bar
$\dot{Q}_C$	Cooling capacity	W
T_C	Curie temperature	K
T_R	Room temperature	K
ΔT_{reg}	Regenerator temperature span	K
$\dot{V}$	Volumetric ow rate	$L\ h^{-1}$

Subscript

2nd	Second-law
C	Cold
f	Fluid
H	Hot
id	Ideal cycle
in	Inlet of the pump
M	Electric motor
max	Maximum value
out	Outlet of the pump
OP	Overall pumping
P	Pumping
R	Room
reg	Regenerator
rem	Remanence
s	Solid
sys	System
visc	Viscous

Abbreviation

AMR	Active magnetic regenerator
COP	Coefficient of performance

Gd	Gadolinium
MCE	Magnetocaloric e_ect
PFA	Peruoroalkoxy alkane
PMMA	Polymethyl methacrylate
POM	Polyacetal
RTD	Resistance temperature detec
UFSC	Federal University of Santa Catarina

1. Introduction

The magnetocaloric effect (MCE) is the thermal response of a magnetic material when subjected to a changing magnetic field. Magnetocaloric refrigeration harvests the MCE in a regenerative thermodynamic cycle to transfer heat from a low-temperature environment to a high-temperature one by means of magnetic work. The main advantage of magnetocaloric refrigeration compared to well-established vapor compression technologies is that the former is based on internally reversible thermodynamic cycles (i.e., Brayton, Stirling, Ericsson), while the latter are based on an internally irreversible cycle (i.e., Reversed Rankine). The MCE is also reversible in most of the known materials used as solid refrigerants in magnetic cooling cycles, including the benchmark material, gadolinium (Gd), and other alloys, especially those exhibiting the second-order phase transition (Nielsen et al., 2010).

Despite the potential benefits of magnetic refrigeration as an emerging cooling technology, its true gains are yet to be fully verified for near room temperature applications. Commercial devices are yet to be developed, and some of the challenges associated with the technology involve (i) the investigation of new magnetocaloric working materials, (ii) new permanent magnet configurations and (iii) the development of new active magnetic regenerator (AMR) geometries for which a substantial knowledge of the conjugate heat transfer and fluid flow in a porous matrix is essential for achieving an optimum design of the cooling system.

A state-of-the-art magnetic refrigeration system is composed basically of an AMR, a magnetic circuit to promote the change in magnetic field and a hydraulic system synchronized with the magnetic field profile. Magnetic refrigerators are classified as reciprocating or rotary. In a reciprocating AMR cooler, there are fewer moving components, but limitations in the operating frequency due to higher inertia. On the other hand, rotary AMRs are able to perform at higher frequencies with multiple regenerators (larger magnetocaloric mass), but at the expense of more complex flow distribution systems. Magnetic refrigerators are also classified according to the magnetic field change generation, which can be performed by three different configurations: a stationary regenerator with a moving magnetic circuit, a stationary magnetic circuit with a moving regenerator or a pulsed field stationary regenerator using an electromagnetic or superconducting coil. Most AMR devices built recently use Gd as the magnetic refrigerant and magnetic circuits based on permanent magnets. Gschneidner and Pecharsky (2008) and Yu et al. (2010) reviewed the magnetic refrigerators and heat pumps built before 2010. Kitanovski et al. (2015) performed a more updated review. Bjørk et al. (2010) performed a review of the magnetic circuits developed for AMR devices.

Magnetic refrigerators are evaluated in terms of the behavior of the cooling capacity, $\dot{Q}_{\mathrm{C}}$, as a function of the regenerator temperature span, ΔT_{reg}, for different operating conditions. The maximum cooling capacity, $\dot{Q}_{\mathrm{C,max}}$, is reached at zero-span (when the hot and cold reservoir temperatures are almost equal, $T_H \approx T_{\mathrm{C}}$), and the maximum regenerator temperature span, $\Delta T_{\mathrm{reg,max}}$, is attained in a no-load test (when $\dot{Q}_{\mathrm{C}} \approx 0$ W).

In the following, the most relevant rotary magnetic refrigerators built using permanent magnets and Gd as the magnetic refrigerant are reviewed. Russek et al. (2010) presented a stationary AMR composed of twelve Gd packed-sphere beds and a rotating 1.4-T magnetic circuit which attained a $\dot{Q}_{\mathrm{C,max}}$ of 844 W at zero-span and 400 W at a ΔT_{reg} of 8.1 K, both at a flow rate of 600 L h^{-1} and an operating frequency of 3.33 Hz. Okamura et al. (2007) developed a rotor-stator magnetic circuit with a 1.1-T magnetic field rotating inside four stationary beds with 4 kg of Gd spheres and obtained a $\dot{Q}_{\mathrm{C,max}}$ of 540 W and a cooling power of 150 W for a ΔT_{reg} of 5.2 K, both at 762 L h^{-1} and 0.4 Hz. Tura and Rowe (2011) constructed a compact AMR consisting of two 1.4-T rotary magnetic circuits and two stationary magnetocaloric beds packed with 110 g of 0.3-mm Gd spheres which generated a no-load $\Delta T_{\mathrm{reg,max}}$ of 29 K and a cooling power of 50 W at a ΔT_{reg} of 10 K at a maximum operating frequency of 4 Hz. A maximum COP of 1.6 was calculated for a cooling capacity of 50 W at a ΔT_{reg} of 2.5 K and a frequency of 1.4 Hz. Engelbrecht et al. (2012) presented a 24 Gd-sphere packed regenerators rotating continuously in a concentric 1.24 T four-pole magnet assembly which demonstrated a zero-span

$\dot{Q}_{\mathrm{C,max}}$ of 1010 W, a no-load $\Delta T_{\mathrm{reg,max}}$ of 25.4 K and a cooling capacity of 100 W at a ΔT_{reg} of 21 K. The device was able to operate at frequencies up to 10 Hz at a ΔT_{reg} of 12.1 K and 200 W cooling capacity (Lozano et al., 2014). Aprea et al. (2014) developed a 1.1-T rotary magnetic refrigerator with eight packed beds of Gd spheres with a total mass of 1.2 kg which generated a no-load $\Delta T_{\mathrm{reg,max}}$ of 13.5 K. The device was able to attain a $\dot{Q}_{\mathrm{C,max}}$ of 200 W at zero-span and a COP of 1.2 was reported for a cooling capacity of 100 W at a ΔT_{reg} of 3.9 K for a volumetric flow rate of 300 L h^{-1} and a frequency of 0.38 Hz (Aprea et al., 2016).

This paper advances the design aspects of a novel rotary-type magnetic cooling prototype developed at the POLO Laboratories of the Federal University of Santa Catarina (UFSC). The device was built based on the experience of previous magnetic refrigerators, offering innovative solutions for the shortcomings found in the others, e.g. by placing the rotary valves only at the hot end to avoid heat generation at the cold end. Experimental results for different operating frequencies, fluid flow rates, hot reservoir temperatures and cooling loads are presented and discussed. The performance of the device is evaluated in terms of the coefficient of performance (COP) and the overall second-law efficiency (η_{2nd}).

2. Novel rotary magnetic refrigerator

The rotary magnetic refrigerator prototype is illustrated in Fig. 1. A rotary magnetic circuit with a stationary regenerator configuration was chosen due to the following characteristics: (i) good potential for operating at high frequencies with low magnetic forces, (ii) large magnetized volume, (iii) continuous pumping of the fluid without leakage, (iv) compactness, and (v) efficient use of the magnets.

The AMR design comprises a stationary ring-type regenerator and a rotor-stator magnetic circuit, as illustrated in Fig. 2. The regenerator is fabricated in acetal resin with 8 pairs of beds packed with approximately 1.7 kg of Gd spheres with diameters between 425 and 600 μm and a porosity of approximately 40.3%. The dimensions of each bed are approximately 80 (length), 10 (height) and 29.7 mm (width). The resulting regenerator parameters are summarized in Table 1. The heat transfer fluid is a 20/80 vv% mixture of automotive antifreeze (ethylene glycol) and distilled water. The antifreeze includes corrosion inhibitors that protect the Gd regenerator and reduce the freezing temperature of the fluid. However, the viscosity of the glycol mixture can be significantly higher than that of pure water, which may increase the pumping losses.

One of the crucial aspects of the new device was the concern with the efficient use of the magnet system. Since the magnet is the largest and most expensive component of the cooling circuit, an optimized and effective magnetic configuration is mandatory. As seen in Fig. 2, the magnetic field generator is a 2-pole rotor-stator magnetic array. Thus, the operating frequency of the AMR is twice that of the rotor. Fig. 3 shows the magnetic flux density norm obtained from numerical simulations (COMSOL Multiphysics®, 2011) using linear demagnetization permanent magnets with remanence $\|\mathbf{B}_{\mathrm{rem}}\| =$ 1.47 T and relative magnetic permeability μ_r = 1.05. The magnetic circuit generates two regions of high magnetic field and two regions of low magnetic field in a 150-mm long, 20-mm tall magnetic gap. The four regions are illustrated in Fig. 3(a) as openings of 45°, each one of them separated by 90°. These angles correspond, respectively, to the hot and cold blow opening angles (times) at the rotary valves and the angle phase between the blows (see Fig. 4).

From the 3D simulation shown in Fig. 3(b), the average high and low magnetic flux densities for the magnetic gap openings of 45° were $\mu_0\overline{H}_{\mathrm{high}} =$ 0.892 T and $\mu_0\overline{H}_{\mathrm{low}} =$ 0.011 T, respectively. The total volume of Nd-Fe-B permanent magnet in the rotor is $V_{\mathrm{magnet}} =$ 1.91 L, and 4 segments of soft magnetic material (iron) were used to conduct the magnetic flux inside the rotor. The stator was designed using laminated electrical steel to avoid eddy losses. The calculated average energy product of the permanent magnet in the rotor from 3D simulations is 365.1 kJ m^{-3} (the maximum energy density is 409.3 kJ m^{-3}). The Λ_{cool} figure of merit of Bjørt et al. (2008) for the actual magnetic circuit was calculated at 0.217 T$^{2/3}$ for opening angles in the magnetic gap of 45° and assuming that the magnet is being used all the time. The magnetic circuit was simulated for magnetization reversal and, since demagnetization of the permanent magnets was not observed, the actual system was built with the high remanence Nd-Fe-B alloy, i.e., *Shinetsu* N52 ($\|\mathbf{B}_{\mathrm{rem}}\| =$ 1.47 T). The magnetic circuit was characterized and validated experimentally by measuring the magnetic flux density profile in the air gap with a transverse Hall probe with an uncertainty of ± 1%. Fig. 4 shows the comparison between the experimental and the 3D numerical simulation profile of the magnetic flux density norm as a function of the angle measured at the center of the magnetic gap (R = 100 mm). More details on the design of the magnetic circuit are described in Ref. (Lozano, 2015).

The flow distribution system is responsible for the alternating fluid flow in the regenerator beds synchronized with the magnetic circuit rotation. A stationary flow system was preferred in order to eliminate fluid leakages in the device. Two rotary valves were placed at the hot end of the system. The valves are mechanically driven by the same shaft of the magnetic circuit. Each valve was designed with eight ports to facilitate the flow distribution and a rotating seal with two openings of 45° to provide a symmetric and continuous fluid flow through the regenerator, thus avoiding over pressure in the hydraulic system. Fig. 4 shows the fluid flow profile generated through a pair of regenerator beds by the rotary valve system as a function of the rotor angle. The outer (static) part of the rotary valves is connected to the hot end flow distributor by PFA hoses. In the hot flow distributor, each channel is bifurcated to service the regenerator pairs. Therefore, at any instant in time the two magnetized bed pairs experience a cold blow, the two demagnetized bed pairs are opened for a hot blow and the four remaining bed pairs are closed for flow. The regenerator ring is embedded into the magnetic gap and connected on each side to flow distributors, as shown in the exploded view of the main components of the AMR device presented in Fig. 5. For sure, a continuous ring-type regenerator with a larger odd number of beds would reduce the demagnetization factor, in addition to the forces and torque fluctuations arising from interactions with the rotating magnet. However, this would require a larger number of magnetic poles and/or a more complex flow distribution system.

A schematic diagram of the hydraulic system is shown in Fig. 6. The central part of the diagram shows four beds; each of them representing the two bed pairs with the same flow profile. The regenerative cycle gives rise a temperature profile so that the left-hand side of the diagram is the hot end while the right-hand side is the cold end. To reduce parasitic losses at the cold end, the majority of the components in the device, including the rotary valves, are placed at the hot end.

The flow distribution system pumps the heat transfer fluid continuously through the system, as indicated by the arrows in Fig. 6. At the particular instant in time depicted in the diagram, the first regenerator bed (13) is considered magnetized and the third regenerator bed (15) is considered demagnetized. A fluid particle leaving the reservoir (1) is then suctioned by the gear pump (2) that increases the pressure of the fluid stream. A needle valve (4) mounted in a flow bypass sets the flow rate, which is measured by an electromagnetic flow meter (5).

The temperature of the fluid entering the regenerator from the hot side, i.e., the so-called hot reservoir temperature (T1), is set by a parallel-plate heat exchanger (6) connected to a temperature-controlled bath (7). The flow passes through a 0.14-mm filter (8) and a purge valve (9) before entering the high-pressure rotary valve (10) at a pressure P1. The valve allows fluid flow to the demagnetized beds (15) via the hot-end flow distributor (12). The inlet and outlet ports of the regenerator are equipped with check valves (18) to direct the flow coming from the rotary valves. The cold end flow distributor (19) receives the cold fluid coming from the regenerator at a temperature T2 and pressure P2. The fluid flow rate at the cold end is measured by a paddle wheel flow meter (21) before entering the screw-type heat exchanger (22) in which an immersion electric heater is used to simulate a thermal load. A flow bypass at the cold end (24) controls the flow rate through the heat exchanger (22). After the thermal load is applied, the flow at a temperature T3 and pressure P3 returns to the magnetized beds (13) which are opened to flow by the low-pressure rotary valve (11). Finally, the fluid flow returns to the reservoir (1) at a temperature T4 and pressure P4.

The absence of a rotary valve at the cold end requires the use of check valves in the channels of the regenerator beds, as shown in the cutaway view of the regenerator, magnetic circuit and cold end flow distributor in Fig. 7. The check valves prevent any flow bypassing the cold heat exchanger. Also, the fluid flow rate is measured at the cold end just before entering the heat exchanger. The hot and cold end flow distributors were fabricated in acrylic resin (PMMA) to allow visual access to the correct functioning of the flow distribution, i.e., movement of the check valves.

The gear pump, driven by a 1.1 kW electrical motor, is capable of giving a maximum differential pressure of 8.7 bar and a maximum flow rate of 1200 kg h^{-1} (for an ethylene-glycol mixture). The system has been designed so that the flow is always unidirectional in the hot and cold heat exchangers. Most of the tubing, connections and fittings are made from stainless steel for its reliability and resistance to corrosion and leakage. Flexible nylon tubing was used in parts of the system to facilitate connections and detect undesired air bubbles.

The magnet assembly and the rotary valve system are driven by a 0.75-kW electrical motor coupled with a helical gear unit reduction which delivers an output torque of 58 Nm at a maximum rotational speed of 124 rpm. The gearmotor is controlled by a frequency inverter. Synchronized pulleys, belt and tensioner avoid slippage due to the high and jittery torque. The pulleys have a ratio of 2:1, which allows for operating AMR frequencies up to 8 Hz. The actual power consumption of the gearmotor driver was measured by a three-phase power transducer located between the main electrical supply and the frequency inverter.

The experimental device is equipped with sensors to measure fluid temperature, absolute pressure, volumetric flow rate, frequency, torque and power. A signal conditioning and data acquisition system operated using Labview is responsible for data logging and storage. The operating frequency and the fluid flow rate are set by frequency inverters controlling the electrical motors. The fluid flow rate can also be controlled manually by a needle valve acting as a flow bypass. The heat load is varied through the voltage input to the immersion heater in the cold heat exchanger. The actual power consumption of the heater was measured by single-phase power transducers located between the main electrical supply and the heater. The hot reservoir temperature, i.e., the temperature of the fluid entering the regenerator from the hot heat exchanger, is controlled by the thermostatic bath.

The accuracies and the ranges of the transducers provided by the manufacturers are summarized in Table 2.

3. Performance metrics

Four different parameters can be varied independently in the experimental setup, namely, the operating frequency, $f = 1/\tau$ (where τ is the cycle time), the volumetric flow fluid rate, $\dot{V}_{\mathrm{f}}$, the hot reservoir temperature, T_{H}, and the thermal load, $\dot{Q}_{\mathrm{C}}$. The performance of AMRs can be evaluated in terms of the utilization, ϕ, defined as the ratio of thermal capacities of the fluid and solid (Rowe et al., 2005):

$$\phi = \frac{m_{\mathrm{f}} c_{\mathrm{f}}}{m_{\mathrm{s}} c_{\mathrm{s}}} \qquad (1)$$

where the mass of the fluid is the product of the mass flow rate and the blow period and c_{f} and c_{s} stand for the specific heat capacities of the fluid and solid matrix, respectively.

The performance of the cooling system can be evaluated in terms of the coefficient of performance (COP), which is defined as the ratio of the cooling capacity to the power consumption of the system,

$$\mathrm{COP} = \frac{\dot{Q}_{\mathrm{C}}}{\dot{W}_{\mathrm{P}} + \dot{W}_{\mathrm{M}}} \qquad (2)$$

where $\dot{W}_{\mathrm{P}}$ and $\dot{W}_{\mathrm{M}}$ are the electrical powers required to drive the fluid pump and the motor that moves the magnet assembly and the rotary valve system, respectively. In this work, the fluid pumping power was calculated based on the viscous dissipation rate, $\dot{W}_{\mathrm{visc}}$, calculated as follows:

$$\dot{W}_{\mathrm{P}} = \frac{\dot{W}_{\mathrm{visc}}}{\eta_{\mathrm{OP}}} = \frac{\dot{V}_{\mathrm{f}} \Delta p_{\mathrm{sys}}}{\eta_{\mathrm{OP}}} = \frac{\dot{V}_{\mathrm{f}} \left(p_{\mathrm{out}} - p_{\mathrm{in}} \right)}{\eta_{\mathrm{OP}}} \quad (3)$$

where Δp_{sys}, p_{out} and p_{in} correspond to the system pressure drop and the measured fluid pressures at the outlet and inlet of the pump, respectively, and η_{OP} is the overall pump efficiency assumed equal to 0.7. The motor power was experimentally measured as the plug power consumed by the frequency inverter.

Another performance metric used to compare different cooling technologies using the same thermodynamic baseline is the second-law efficiency, η_{2nd} Hermes and Barbosa (2012). It is defined as the ratio of the power consumed by the actual device to that of a reversible system operating within the same temperature span associated with the hot and cold environments (ΔT_{sys}). In terms of the COPs, it is given by:

$$\eta_{\mathrm{2nd}} = \frac{\mathrm{COP}}{\mathrm{COP}_{\mathrm{id}}} \quad (4)$$

where $\mathrm{COP}_{\mathrm{id}}$ is the COP of an ideal (Carnot), i.e. totally reversible, refrigerator, which is defined in terms of the hot and cold environment temperatures, T_{H} and T_{C}, as follows:

$$\mathrm{COP}_{\mathrm{id}} = \frac{T_{\mathrm{C}}}{T_{\mathrm{H}} - T_{\mathrm{C}}} \quad (5)$$

A detailed performance analysis of a state-of-the-art magnetic refrigerator is presented in Lozano et al. (2013), where the maximum values of η_{2nd} were around 5%. The experiments performed in this work were also evaluated in terms of the actual COP and η_{2nd}. Further evaluation of the thermodynamic performance of this device are reported in Capovilla et al. (2016).

4. Results and discussions

The experimental setup allows mapping the four different parameters which influence the performance of the magnetic refrigerator: (i) operating frequency (f), (ii) volumetric flow rate (

$\dot{V}_{\mathrm{f}}$), (iii) hot reservoir temperature (T_{H}) and (iv) thermal load ($\dot{Q}_{\mathrm{C}}$). The system performance was evaluated independently for each of the four parameters. The volumetric fluid flow rate reported here was measured by the cold end flow meter to account for possible internal leakages in the flow distribution system or in the regenerator beds. Also, the flow rate measured by this flow meter is the actual flow rate passing through the cold heat exchanger. In all experiments, the room temperature was maintained at 293 ± 1 K by the room air conditioning system.

The system pressure drop (Δp_{sys}) is experimentally evaluated as the average difference between the pressure of the fluid entering the hot end of the regenerator and the pressure of the fluid exiting the hot end of the regenerator. As the assessment of the performance of the hot and cold heat exchangers was outside the scope of the present paper, the hot and the cold environment temperatures, T_{H} and T_{C}, are defined as the time-average temperature of the fluid entering the hot and the cold ends of the regenerator, respectively. The regenerator temperature span (ΔT_{reg}) is experimentally assessed as the average difference between the temperature of the fluid exiting the hot end of the regenerator and the temperature of the fluid exiting the cold end of the regenerator. The experiment was considered to have reached steady state when the standard deviation of the regenerator temperature span remained below 0.03 K during a sampling period of 120 s ($\sigma_{\Delta T_{\mathrm{reg}}} <$ 0.03 K). This value is much lower than the expanded uncertainty of a resistance temperature detector (RTD) at room temperature, which is approximately $U_{T_{\mathrm{f}},95} \sim$ 0.22 K. The original sampling frequency of the experimental measurements was 1 kHz, but since the transient physical phenomena in the experiment are not high-frequency, the samples are averaged and recorded every 0.5 s (sampling rate 2 Hz), for a better representation of the measured parameters and to eliminate any high-frequency noise. The sampling period is 120 s, which results in a number of measurements $N \simeq 240$ per experiment. The result for each parameter corresponds to the time-averaged value of the measured samples for each experiment and the expanded uncertainties (expressed for a 95% level of confidence) for those results are presented as error bars at the plots. The uncertainty propagation analysis to determine the expanded uncertainties of the measured and calculated parameters was carried out according to the procedures of Moffat (1988) and Coleman and Steele (2009). The complete analysis is available in Ref. (Lozano, 2015).

Fig. 8 shows the system pressure drop for the packed regenerator at a frequency of 0.8 Hz as a function of the fluid flow rate measured by the cold end flow meter. Both the experimental absolute pressures and the volumetric flow rate show some variation between beds. This can be related to variations in packing structure between adjacent beds. The expanded uncertainty for such differences is included in Fig. 8 as error bars. The volumetric flow rate measured at the cold end flow meter was, on average, 4.96% lower than the volumetric flow rate measured at the hot end flow meter. This can be attributed to fluid internal leakages in the flow distribution system and within the regenerator beds before reaching the cold heat exchanger.

Performance curves at a constant AMR operating frequency of 0.4 Hz and a hot reservoir temperature of approximately 295.7 K ($T_{\mathrm{bath}} =$ 295.15 K) are shown in Fig. 9. The evaluation of the physical properties of the Gd spheres revealed a Curie temperature of approximately 290 K, so as a rule of thumb a better performance of the machine can be expected when the temperature at the center plane of the regenerator approaches this temperature. In the experiments, volumetric flow rates of 100, 125 and 150 L h^{-1} (ϕ = 0.47, 0.58 and 0.70, respectively) were held constant while the thermal load was increased gradually until steady state was attained at the desired $\dot{Q}_{\mathrm{C}}$. At 150 L h^{-1} (ϕ = 0.70), the system was capable of providing a cooling capacity of 62.5 W while holding a regenerator temperature span of 6.1 K. The performance curves in Fig. 9 are in good agreement with those normally found in the literature. For an AMR working at low temperature spans around the Curie temperature of the magnetocaloric material, the cooling capacity is usually large because the MCE of the material is also large for both magnetization and demagnetization at these temperatures. On the other hand, if a large temperature span is desired, the cooling capacity will be reduced because the MCE is likely to be small at the temperatures corresponding to the ends of the regenerator. In theory, the magnetic circuit operates approximately at room temperature (~ 293 K). At the conditions of very low temperature span in Fig. 9, it may be possible that the regenerator was slightly warmer than the magnetic circuit and the surrounding air and, as a result, heat transfer from the regenerator to its surroundings could have contributed to the increase in cooling capacity at very small temperature spans (see, for instance, the performance curve for 125 L h^{-1}). The occurrence of slightly negative temperature spans at the highest cooling capacities for 100 and 150 L h^{-1} is due to the difficulty in obtaining a value of ΔT_{reg} exactly equal to zero when the independent variable is the thermal load.

The dependence of the no-load regenerator temperature span on the utilization (ϕ) was investigated for two conditions: (i) fixed volumetric flow rate and (ii) fixed operating frequency, both for a constant hot reservoir temperature of approximately 295.7 K (Fig. 10). In the former scenario, the flow rate was held at 150 L h^{-1} and the operating frequency was varied from 0.2 to 1.4 Hz (ϕ = 1.43 to 0.20, respectively). A peak in the regenerator temperature span of 12.0 K was found at 1 Hz (ϕ = 0.28). In the latter case, the operating frequency was held fixed at 0.8 Hz and the fluid flow rate was varied from 50 to 175 L h^{-1} (ϕ = 0.12 to 0.40) and a peak on ΔT_{reg} of 11.7 K was found at ϕ = 0.40.

The influence of the hot reservoir temperature on the regenerator temperature span for an applied thermal load of 60 W was evaluated at an operating frequency of 0.8 Hz and different volumetric flow rates of 150, 175 and 200 L h^{-1} (ϕ = 0.35, 0.41 and 0.47, respectively), as seen in Fig. 11. The hot reservoir temperature was varied gradually from approximately 290.2 to 301.8 K (T_{bath} from 289.15 to 301.15 K, respectively). The curves resemble that for the MCE of Gd, but with peaks at higher temperatures. This is probably due to the maximization of the MCE over the entire regenerator as the average temperature between the hot and cold ends approached the Curie temperature of Gd (Trevizoli et al., 2011). For instance, the peak of the regenerator temperature span was 8.2 K for a volumetric flow rate of 200 L h^{-1} and a hot reservoir temperature of 296.1 K.

Fig. 12 shows the performance curves for an operating frequency of 0.8 Hz, a hot reservoir temperature of approximately 294.1 K and different volumetric flow rates of 150, 175 and 200 L h^{-1} (ϕ = 0.35, 0.41 and 0.47, respectively). Due to pressure drop limitations, the fluid flow rate was not raised above 200 L h^{-1}. Like the results in Fig. 9, the regenerator temperature span was observed to increase with the flow rate irrespective of the cooling capacity. The maximum no-load ΔT_{reg} was 10.6 K, while the maximum zero-span cooling capacity was 150.0 W, both at 200 L h^{-1}. At a cooling capacity of 80.4 W the regenerator was able to sustain a temperature span of 7.1 K, also at 200 L h^{-1}. It is important to point out that the interpolated maximum system capacity ($\Delta T_{\mathrm{sys}} = 0$) was approximately 139.8 W.

The thermodynamic performance of the device was evaluated in terms of the coefficient of performance (COP) and the overall second-law efficiency (η_{2nd}). The actual COP is calculated based on Eq. (2), where $\dot{Q}_{C}$ corresponds to the thermal load imposed by the electric heater at the cold heat exchanger and $\dot{W}_{M}$ is the plug power of the electrical motor. These two parameters are experimentally measured by the power transducers. Conversely, due to the manual bypassing of the fluid to set the volumetric flow rate, the fluid pumping power, $\dot{W}_{P}$, is calculated based on Eq. (3).

Fig. 13 shows the calculated COP as a function of the regenerator temperature span for several selected thermal loads and positive-span experiments. It can be inferred that the actual COP increases at higher thermal loads but at the expense of lower temperature spans. The average motor power for experiments at 0.4 and 0.8 Hz were 88.8 and 106.4 W, respectively. The average calculated pumping power for experiments at 150 and 200 L h^{-1} were 19.5 and 43.3 W, respectively. For instance, an experiment with an operating frequency of 0.8 Hz, a fluid flow rate of 175 L h^{-1}, a T_{H} of 294.1 K, a $\dot{Q}_{C}$ of 81.0 W produced over a ΔT_{reg} of 6.5 K and a COP of 0.58.

The overall second law efficiency is calculated as the ratio of the actual and ideal COP using Eq. 4. Fig. 14 shows the results for the calculated η_{2nd} as a function of the regenerator temperature span for the experimental results presented in Fig. 13. η_{2nd} accounts for all losses and quantifies the actual performance of the system in comparison with an ideal (Carnot) system operating between the same environment temperatures. The maximum value of η_{2nd} obtained for the present system was 1.16% for an experiment at an operating frequency of 0.8 Hz, fluid flow rate of 200 L h^{-1}, cooling power of 80.4 W, T_{H} of 294.1 K and ΔT_{reg} of 7.1 K, where the calculated COP and COP_{id} were 0.54 and 45.99, respectively. The magnitude of the actual pump work is similar in magnitude to the calculated fluid pumping power ($\dot{W}_{P}$) for each experimental condition. Therefore, the combined uncertainty for the calculated COP, which includes the combined uncertainty of $\dot{W}_{P}$, does not change significantly the uncertainty of the actual COP and η_{2nd}, represented as vertical error bars in Figs. 13 and 14, respectively.

A summary of selected experimental results generated in this work and their calculated performance is presented in Table 3.

5. Conclusions

A novel rotary magnetic refrigerator with a 2-pole rotor-stator magnetic circuit and a stationary 8 pairs bed regenerator packed with 1.7 kg of Gd spheres was designed and built at the Federal University of Santa Catarina. The magnetic circuit was designed aiming for an efficient use of the permanent magnets and a flow distribution system was specially designed to be positioned at the hot end. The effectiveness of both the magnetic circuit and flow distribution system were validated experimentally. The first experimental analyses of the magnetic refrigerator resulted in a maximum cooling capacity of 150 W at zero temperature span, 0.8 Hz, 200 L h^{-1} ($\phi = 0.47$) and $T_H = 294.1$ K. The maximum temperature span attained in the AMR was 12 K in a zero-load experiment, 1 Hz, 150 L h^{-1} ($\phi = 0.28$) and $T_H = 295.7$ K. The device was able to absorb 80.4 W while maintaining a temperature span of 7.1 K at an operating frequency of 0.8 Hz, a volumetric flow rate of 150 L h^{-1} ($\phi = 0.47$) and a hot reservoir temperature of 294.1 K. For these operating conditions, the calculated COP was 0.54 and the maximum calculated η_{2nd} was 1.16%. The performance results are yet modest in comparison with established cooling technologies or with some of the state-of-the-art magnetic refrigerators. An increase in the experimental performance and efficiency is expected at higher fluid flow rates and operating frequencies. It has been demonstrated that there is still room for improvement, especially regarding the regenerator configuration, which will be the subject of further studies.

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Figure 1: Photograph of the novel rotary magnetic refrigerator developed at POLO/UFSC.

Figure 2: Cross section view of the magnetic circuit and the stationary annular regenerator. The black arrows indicate the direction of the remanence of the Nd-Fe-B permanent magnet segments.

Figure 3: (a) 2D simulation of the magnetic flux density norm of the circuit using permanent magnets with $\| B_{\mathrm{rem}} \| =$ 1.47 T. (b) 3D simulation of the magnetic flux density norm in one quarter of the circuit (mid-plane at the center of the magnet). Arrows indicate the direction of the remanence.

Figure 4: Experimental and simulated magnetic flux density and normalized fluid flow profiles as a function of the rotor angle.

Figure 5: Exploded view of the main components of the rotary device.

Figure 6: Schematic diagram of the hydraulic system: (1) reservoir, (2) pump, (3) relief valve, (4) needle valve (flow bypass), (5) electromagnetic flow meter, (6) hot heat exchanger, (7) temperature controlled bath, (8) filter, (9) purge valve, (10) high-pressure rotary valve, (11) low-pressure rotary valve, (12) hot end flow distributor (13-16) regenerator beds, (17) magnet, (18) check valves, (19) cold end flow distributor, (20) filter, (21) paddle wheel flow meter, (22) cold heat exchanger, (23) purge valve, and (24) flow bypass.

Figure 7: Cutaway view of the regenerator, magnetic circuit and cold end flow distributor.

Figure 8: System pressure drop for the system operating at a frequency of 0.8 Hz.

Figure 9: Performance curves for different volumetric flow rates at an operating frequency of 0.4 Hz and hot reservoir temperature of approximately 295.7 K.

Figure 10: No-load regenerator temperature span dependence on the utilization when the frequency and flow rate are varied for a hot reservoir temperature of approximately 295.7 K.

Figure 11: Regenerator temperature span as a function of the hot reservoir temperature for a thermal load of 60 W, operating frequency of 0.8 Hz and volumetric flow rates of 150, 175 and 200 L h^{-1} (ϕ = 0.35, 0.41 and 0.47, respectively).

Figure 12: Performance curves for different volumetric flow rates at an operating frequency of 0.8 Hz and an average hot reservoir temperature of approximately 294.1 K.

Figure 13: Calculated COP results as a function of the temperature span for some of the non-zero thermal load and positive temperature span experiments.

Figure 14: Calculated η_{2nd} as a function of the corresponding regenerator temperature span for the experimental results presented in Fig. 13.

Table 1: Regenerator parameters.

Parameter	Symbol	Value	Unit
Number of beds	n_{bed}	16	-
Length of the beds	L_{bed}	80	mm
Bed cross section area	A_c	280	mm^2

Regenerator volume	V_{reg}	0.36	L
Total regenerator mass	m_{reg}	1.7	kg
Regenerator porosity	ε	0.40	-
Magnetocaloric material	Gd	Gadolinium	-
Spheres diameter	d_{p}	0.425-0.6	mm
Regenerator casing	POM	Polyacetal	-

Table 2: Accuracies and ranges of the transducers provided by the manufacturers.

Sensor	Accuracy	Range
Temperature (RTD Pt-100)	$\pm(0.002T - 0.396)$ K	$273 - 373$ K
Temperature (Thermocouple)	±1 K	$233 - 398$ K
Absolute pressure	±0.016 bar	$0 - 16$ bar
Hot end fluid flow rate	$\pm0.5\%$ of rate	$0 - 1200$ L h^{-1}
Cold end fluid flow rate	±11.3 L h^{-1}	$9 - 1134$ L h^{-1}
Single-phase power transducer	±2.5 W	$0 - 1000$ W
Three-phase power transducer	±8.2 W	$0 - 3290.9$ W
Torque transducer	$\pm0.3\%$ of torque	$0 - 20$ Nm

Table 3: Summary of selected experimental results and their calculated performance.

f	$\dot{V}_f$	ϕ	T_H	$\dot{Q}_C$	ΔT_{reg}	$\dot{W}_M$	$\dot{W}_P$	COP	η_{2nd}
(Hz)	(Lh^{-1})	(−)	(K)	(W)	(K)	(W)	(W)	(−)	(%)
1.4	151.7	0.20	296.0	61.6	7.1	145.0	20.7	0.37	0.80
0.8	174.2	0.41	296.0	61.0	7.9	108.1	29.3	0.44	1.12
0.8	201.5	0.47	294.2	80.4	7.1	106.5	43.7	0.54	1.16
0.4	150.0	0.70	295.8	62.5	6.1	89.5	19.1	0.58	1.07
0.8	176.6	0.41	294.1	81.0	6.5	107.5	31.1	0.58	1.12
0.8	151.7	0.36	294.0	81.2	5.0	103.8	20.7	0.65	0.88
0.4	150.5	0.70	295.9	80.8	3.7	87.6	18.6	0.76	0.73
0.8	200.4	0.47	294.3	120.4	3.7	103.0	42.1	0.83	0.71